



HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
EEE 127: Electronic Devices and Circuits

Batch: 26th
Date: 01/01/2025

Midterm Examination

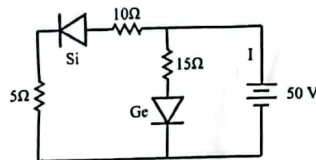
Semester: Fall 2024
Time: 1 Hour 30 minutes
Total Marks: 30

5 × 6 = 30

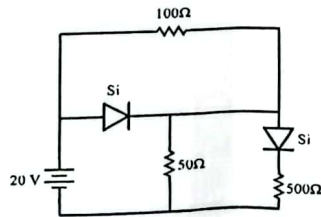
Answer any five questions of the following.

1. (a) Explain band theory with the necessary diagrams. CLO1 3
(b) What is the significance of the energy gap (band gap) in semiconductor materials? How does it affect the electrical properties of the material? CLO1 3

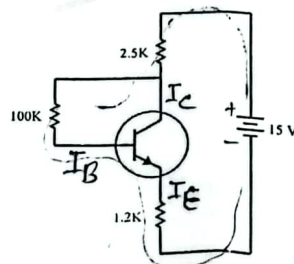
2. (a) Define knee voltage and reverse breakdown voltage for a diode. Explain their significance in diode operation. CLO1 2
(b) Compare silicon (Si) and germanium (Ge) diodes. Why is silicon preferred in most electronic applications? CLO2 2
(c) Determine the Current I from the following circuit: CLO2 2



3. (a) Sketch and explain the V-I characteristics of a p-n junction diode, highlighting key regions of operation. CLO1 2
(b) Determine the Current I_{100} and I_{500} from the following diagram: CLO2 4

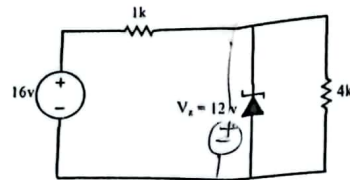


4. (a) Explain how a BJT can be used as a switch in electronic circuits. CLO1 2
(b) Determine the I_E , I_B , I_C , V_E , V_B , V_C , V_{BE} , V_{CE} , V_{BC} from the following circuit. CLO2 4

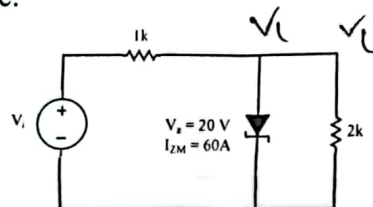


$$\beta = 100$$

5. (a) Write the relationship equations among α , β and γ in transistor? Explain with proper figure. CLO1 2
- (b) Compare the input and output characteristics of CB, CE, and CC configurations in transistor. CLO1 4
6. (a) Explain the working principle of a Bipolar Junction Transistor. CLO1 3
- (b) Determine the current across Zener diode and current across the load resistor. CLO2 3



7. (a) How does a Zener diode differ from a regular diode in terms of operation and application? CLO1 3
- 6 (b) Determine the range of values of V_i that will maintain the Zener diode in "ON" state. CLO2 3





Answer any 3 (Three) from the following sets of questions

[The figures in the right margin indicate full marks. Symbols used have their usual meaning.]

Sl.No.	Questions	CLOs	Marks
✓	(a) Define homogeneous linear system.	CLO1	01
	(b) Solve the following system of linear equations by row echelon form: $\begin{aligned} x_1 + x_2 + 2x_3 &= 8 \\ -x_1 - 2x_2 + 3x_3 &= 1 \\ 3x_1 - 7x_2 + 4x_3 &= 10 \end{aligned}$	CLO3	03
	(c) Solve the following homogeneous system of linear equations by using Gauss-Jordan elimination method. $\begin{aligned} 2x_1 + 2x_2 - x_3 + x_5 &= 0 \\ -x_1 - x_2 + 2x_3 - 3x_4 + x_5 &= 0 \\ x_1 + x_2 - 2x_3 - x_5 &= 0 \\ x_3 + x_4 + x_5 &= 0 \end{aligned}$	CLO3	06
✓	(a) What is a matrix?	CLO1	01
	(b) Find the solution of linear system by using inverse matrix, the following equations: $\begin{aligned} x_1 + 2x_2 + 3x_3 &= 5 \\ 2x_1 + 5x_2 + 3x_3 &= 3 \\ x_1 + 8x_3 &= 17 \end{aligned}$	CLO3	03
	(c) Find the inverse of the following matrix by using row canonical form: $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 3 \\ 1 & 0 & 8 \end{bmatrix}$	CLO3	06
3.	(a) Define linear dependence and independence.	CLO1	01
	(b) Show that the set of vectors $\{(2, -1, 4), (3, 6, 2), (2, 10, -4)\}$ are linearly independent.	CLO2	04
	(c) Prove that the vectors $(1, 2, 0), (0, 5, 7)$ and $(-1, 1, 3)$ form a basis for R^3 .	CLO3	05
4.	(a) What is Hermitian matrix?	CLO1	01
	(b) Using Cayley-Hamilton theorem find the inverse of the matrix $A = \begin{bmatrix} 1 & 2 & 2 \\ 3 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$.	CLO2	03
	(c) Show that every square matrix can be uniquely expressed as $P + iQ$ where P and Q are Hermitian matrix.	CLO3	06
5.	(a) Define rank of a matrix.	CLO1	01
	(b) Find the rank, ImT, nullity and KerT of the following linear transformations: $T: R^3 \rightarrow R^3$ and $T(x, y, z) = (x + 2y - z, y - x, x - z)$.	CLO3	03
	(c) Find the Eigenvalues and Eigenvectors of the following matrix: $A = \begin{bmatrix} 4 & 6 & 6 \\ 1 & 3 & 2 \\ -1 & -4 & -3 \end{bmatrix}$	CLO3	06

Hamdard University Bangladesh

Faculty of Science, Engineering and Technology

Midterm Examination (Fall'24)

PHY 111 Electricity & Magnetism/ PHY 121 Physics II/

PHY 121 Physics II (Mechanics, Electricity, Magnetism and Modern Physics)

Time: 1.5 hours

Marks: 30

Answer any 3 (Three) of the following sets of questions

- 1 (a) State and explain the Coulomb's law of electrostatics. 2 CLO1
(b) Calculate the electrical field due to an electric dipole at a point on the perpendicular bisector of the dipole with the help of neat and clean figure. 4 CLO2
(c) Four equal point charges, $+3\text{mC}$, are placed at the four corners of a square that is 40 cm on a side. Find the force on any one of the charges by drawing the figure. 4 CLO3
- $\mu = 10^{-6}$
 $8.58 \times 10^5 \text{ N}$
- 2 (a) State Gauss law of electrostatics and write down the formula. 2 CLO1
(b) Derive the electric field equation due to a line of charge applying the Gauss law. 4 CLO2
(c) Portions of two large sheets of charge with uniform surface charge densities $S_+ = +6.8 \text{ mC/m}^2$ and $S_- = -4.3 \text{ mC/m}^2$ are shown in Figure 2(b) Find the electric field, E (a) to the left of the sheets, (b) between the sheets, and (c) to the right of the sheets. 4 CLO3

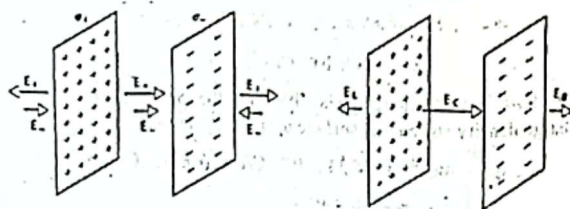


Figure 2(b)

- 3 (a) Define electric potential and electric potential energy. 2 CLO1
(b) Derive the electric potential due to a dipole where two equal charges of opposite sign separated by a small distance, $2a$ and also explain each special cases of electric potential. 4 CLO2
(c) What is the electric potential energy of the charge configuration in the figure below? Given, $q_1 = +1 \times 10^{-8} \text{ C}$, $q_2 = -2 \times 10^{-8} \text{ C}$, $q_3 = +3 \times 10^{-8} \text{ C}$, $q_4 = +2 \times 10^{-8} \text{ C}$, and $a = 1 \text{ m}$. 4 CLO3

$7.4272 \times 10^{-6} \text{ J}$

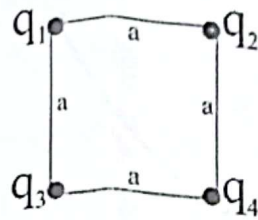


Figure-3(c)

- 4 (a) What is a capacitor? What do you mean by the charge of a capacitor? 2 CLO1
- (b) Derive the equation of capacitance of a parallel plate capacitor with the help of proper figure. 4 CLO2
- (c) How much charge is stored in a capacitor consisting of two concentric spheres of radii 30 and 31 cm if the potential difference is 500V? 4 CLO3

$$5.1685 \times 10^{-7} \text{ C}$$

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 121: Structured Programming Language

Batch: 26th

Semester: Fall 2024

Date: 07/01/2025

Midterm Examination

Time: 1 Hour 30 Minutes

Total Marks: 30

Answer any **five** questions.

$5 \times 6 = 30$

- | | |
|---|---|
| <p>1. (a) Define the following:</p> <ul style="list-style-type: none"> i. Programming Language ii. Syntax iii. Language Processor <p>(b) Differentiate between Compiler and Interpreter.</p> | <p>CLO1 $3 \times 1 = 3$</p> <p>CLO3 3</p> |
| <p>2. (a) Briefly describe the characteristics of a good algorithm.</p> <p>(b) Write the algorithm and draw the corresponding flowchart to determine the smaller number between two given numbers (<i>num1</i> and <i>num2</i>).</p> | <p>CLO2 3</p> <p>CLO2 3</p> |
| <p>3. (a) Why is the <code>main()</code> function essential in every C program?</p> <p>(b) Find errors, if any, in the following programs —</p> <ul style="list-style-type: none"> i. <pre>#include <stdio.h> int main() { int a, b; printf("Enter two numbers: "); scanf("%d %d", a, b); printf("The sum is: %d\n", a + b); return 0; }</pre> ii. <pre>#include <stdio.h> int main() { char c = 'A' printf("The character is: %c\n", c); return 0; }</pre> iii. <pre>#include <stdio.h> int main() { int a, b; printf("Enter two numbers: "); scanf("%d %d", a, b); product = a * b; printf("The product is: %d", product); return 0; }</pre> | <p>CLO4 2</p> <p>CLO5 $4 \times 1 = 4$</p> |



```
iv    #include <stdio.h>
      void main() {
          printf("Hello, World!");
          return 0;
      }
```

4. (a) What are the rules for naming an identifier in C? CLO4 2
(b) Which of the following are invalid variable names and why? CLO4 4×.5=2
i. *Column-total*
ii. *1st_row*
iii. *Maximum*
iv. *Float*
(c) How does the double data type differ from float in terms of precision and storage? CLO3 2

5. (a) Briefly explain Implicit and Explicit type conversion with examples. CLO3 3
(b) Find out the output of the following code snippet — CLO4 3

```
#include <stdio.h>
main() {
    int a, b, c, d;
    a = 12;
    b = 8;
    c = ++a - b;
    printf("a = %d b = %d c = %d\n", a, b, c);
    d = b++ + a;
    printf("a = %d b = %d d = %d\n", a, b, d);
    printf("a += b = %d\n", a+=b);
    printf("%d\n", (c<d) ? 1 : 0);
}
```

6. (a) What does the format specifier **%7.2f** mean in printf()? CLO4 2
(b) Write the following programs in C — CLO4 2×2=4
i. Program to check if a given character is vowel or not.
ii. Program to swap the values of two integer variables.

7. Write a C program that implements a simple calculator using a switch-case statement. CLO4 6

Requirements:

- Input two numbers (num1 and num2) using scanf.
- Input an operation (+, -, *, or /).
- Perform the selected operation using a switch-case and display the result.
- Handle division by zero.
- Handle false input.



Department of Computer Science and Engineering
Faculty of Science and Engineering
Bachelor of Science in Computer Science and Engineering
ENG127-English Skills
Midterm Examination
Fall 2024

26th Batch
13th January 2025

2nd Semester
Time: 1 hour 30 minutes

Total Marks: 30

CLOs	Level	Marks
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Section A

Read the following passage and answer questions 1 - 3:

Asia is a vast and diverse continent, encompassing a wide range of climates from the frigid tundras of Siberia to the tropical rainforests of Southeast Asia. The geographic position of Asia influences its climate due to factors like elevation, ocean currents, and the vast landmass.

In the northern parts of Asia, including Siberia, the climate is predominantly land-based, characterized by long, cold winters and short, warm summers. This is due to its high geographic position and distance from large bodies of water, which limits the moderating effects of the ocean. The temperature can drop drastically in winter, often reaching below -30°C in some areas.

In Central Asia, the climate is primarily desert and semi-arid. The region is influenced by the rain shadow effect of the Himalayas, which blocks moist oceanic winds from reaching the interior. This results in low precipitation and extreme temperature variations between day and night. Summers are hot, with temperatures often exceeding 40°C, while winters are cold and dry.

Southeast Asia, including countries like Thailand, Vietnam, and the Philippines, experiences a tropical climate with high temperatures and high humidity year-round. The region is affected by monsoon winds, which bring heavy rainfall from May to October. This monsoon season is crucial for agriculture, as it provides the necessary water for rice cultivation and other crops.

In East Asia, including China, Japan, and Korea, the climate varies from humid subtropical in the south to temperate in the north. The region experiences distinct seasonal changes with hot, humid summers and cold, dry winters. The monsoon season also affects East Asia, bringing heavy rain to much of the region during the summer months.

Overall, the climate in Asia is highly varied, shaped by a complex interplay of geographic and meteorological factors!

1. Choose the correct answer from the following alternatives:

CLO3	Understand	5x1=
------	------------	------

a. What type of climate is predominant in northern Asia, including Siberia?

- | | | | |
|-------------|------------|-----------------|-----------------|
| i. Tropical | ii. Desert | iii. Land-based | iv. Subtropical |
|-------------|------------|-----------------|-----------------|

b. Which climatic factor heavily influences Central Asia's weather?

- | | |
|-------------------|-------------------------|
| i. Ocean currents | ii. Geographic position |
| iii. Elevation | iv. Rain shadow effect |

c. What type of climate does Southeast Asia experience?

- | | | | |
|----------------|------------|---------------|---------------|
| i. Continental | ii. Desert | iii. Tropical | iv. Temperate |
|----------------|------------|---------------|---------------|

d. Which region in Asia experiences a humid subtropical climate?

- i. Northern Asia ii. Central Asia iii. Southeast Asia iv. East Asia

e. What season brings heavy rainfall to Southeast Asia?

- i. Winter ii. Spring iii. Monsoon iv. Summer

2. Answer the following questions:

- a. What type of climate is predominant in northern Asia?
b. How does the rain shadow effect influence Central Asia's climate?
c. What type of climate does Southeast Asia experience?
d. Which region in Asia features a humid subtropical climate?
e. How do ocean currents and wind patterns affect the climate in different parts of Asia?

Analyze 5x2=10

3. Fill in the blanks with suitable words:

- a. The geographical position of Asia influences its climate due to factors like _____, ocean currents, and the vast _____.
b. In the northern parts of Asia, the climate is predominantly _____, characterized by long, cold winters and short, warm summers.
c. Central Asia has a _____ climate, influenced by the _____ effect of the Himalayas.
d. Southeast Asia experiences a _____ climate with high temperatures and heavy Monsoon rains from May to October.
e. In East Asia, the climate varies from _____ subtropical in the south to _____ in the north.

Apply 5x1=5

Section B

CLO 1 Understand 2x5=10

3. Answer any two of the followings:

- a) How to improve speaking skill? Explain.
b) Define skimming and scanning, and explain how these techniques can support our progress in enhancing our reading skill.
c) Briefly describe the differences between productive and receptive skills.

above